

```
In [1]: pip install pandas seaborn matplotlib gradio ollama
```

Requirement already satisfied: pandas in c:\users\suras\anaconda2\lib\site-packages (2.2.2)

Requirement already satisfied: seaborn in c:\users\suras\anaconda2\lib\site-packages (0.13.2)

Requirement already satisfied: matplotlib in c:\users\suras\anaconda2\lib\site-packages (3.9.2)

Requirement already satisfied: gradio in c:\users\suras\anaconda2\lib\site-packages (5.37.0)

Requirement already satisfied: ollama in c:\users\suras\anaconda2\lib\site-packages (0.5.1)

Requirement already satisfied: numpy>=1.26.0 in c:\users\suras\anaconda2\lib\site-packages (from pandas) (1.26.4)

Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\suras\anaconda2\lib\site-packages (from pandas) (2.9.0.post0)

Requirement already satisfied: pytz>=2020.1 in c:\users\suras\anaconda2\lib\site-packages (from pandas) (2024.1)

Requirement already satisfied: tzdata>=2022.7 in c:\users\suras\anaconda2\lib\site-packages (from pandas) (2023.3)

Requirement already satisfied: contourpy>=1.0.1 in c:\users\suras\anaconda2\lib\site-packages (from matplotlib) (1.2.0)

Requirement already satisfied: cycler>=0.10 in c:\users\suras\anaconda2\lib\site-packages (from matplotlib) (0.11.0)

Requirement already satisfied: fonttools>=4.22.0 in c:\users\suras\anaconda2\lib\site-packages (from matplotlib) (4.51.0)

Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\suras\anaconda2\lib\site-packages (from matplotlib) (1.4.4)

Requirement already satisfied: packaging>=20.0 in c:\users\suras\anaconda2\lib\site-packages (from matplotlib) (24.1)

Requirement already satisfied: pillow>=8 in c:\users\suras\anaconda2\lib\site-packages (from matplotlib) (10.4.0)

Requirement already satisfied: pyparsing>=2.3.1 in c:\users\suras\anaconda2\lib\site-packages (from matplotlib) (3.1.2)

Requirement already satisfied: aiofiles<25.0,>=22.0 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (24.1.0)

Requirement already satisfied: anyio<5.0,>=3.0 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (4.2.0)

Requirement already satisfied: brotli>=1.1.0 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (1.1.0)

Requirement already satisfied: fastapi<1.0,>=0.115.2 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (0.116.1)

Requirement already satisfied: ffmpeg in c:\users\suras\anaconda2\lib\site-packages (from gradio) (0.6.0)

Requirement already satisfied: gradio-client==1.10.4 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (1.10.4)

Requirement already satisfied: groovy~=0.1 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (0.1.2)

Requirement already satisfied: httpx<1.0,>=0.24.1 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (0.27.0)

Requirement already satisfied: huggingface-hub>=0.28.1 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (0.33.4)

Requirement already satisfied: jinja2<4.0 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (3.1.4)

Requirement already satisfied: markupsafe<4.0,>=2.0 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (2.1.3)

Requirement already satisfied: orjson~=3.0 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (3.10.18)

Requirement already satisfied: pydantic<2.12,>=2.0 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (2.11.7)

Requirement already satisfied: pydub in c:\users\suras\anaconda2\lib\site-packages (from gradio) (0.25.1)

Requirement already satisfied: python-multipart>=0.0.18 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (0.0.20)

Requirement already satisfied: pyyaml<7.0,>=5.0 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (6.0.1)

Requirement already satisfied: ruff>=0.9.3 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (0.12.3)

Requirement already satisfied: safehttpx<0.2.0,>=0.1.6 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (0.1.6)

Requirement already satisfied: semantic-version~=2.0 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (2.10.0)

Requirement already satisfied: starlette<1.0,>=0.40.0 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (0.47.1)

Requirement already satisfied: tomlkit<0.14.0,>=0.12.0 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (0.13.3)

Requirement already satisfied: typer<1.0,>=0.12 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (0.16.0)

Requirement already satisfied: typing-extensions~=4.0 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (4.14.1)

Requirement already satisfied: uvicorn>=0.14.0 in c:\users\suras\anaconda2\lib\site-packages (from gradio) (0.35.0)

Requirement already satisfied: fsspec in c:\users\suras\anaconda2\lib\site-packages (from gradio-client==1.10.4->gradio) (2024.6.1)

Requirement already satisfied: websockets<16.0,>=10.0 in c:\users\suras\anaconda2\lib\site-packages (from gradio-client==1.10.4->gradio) (10.4)

Requirement already satisfied: idna>=2.8 in c:\users\suras\anaconda2\lib\site-packages (from anyio<5.0,>=3.0->gradio) (3.7)

Requirement already satisfied: sniffio>=1.1 in c:\users\suras\anaconda2\lib\site-packages (from anyio<5.0,>=3.0->gradio) (1.3.0)

Requirement already satisfied: certifi in c:\users\suras\anaconda2\lib\site-packages (from httpx<1.0,>=0.24.1->gradio) (2024.8.30)

Requirement already satisfied: httpcore==1.* in c:\users\suras\anaconda2\lib\site-packages (from httpx<1.0,>=0.24.1->gradio) (1.0.2)

Requirement already satisfied: h11<0.15,>=0.13 in c:\users\suras\anaconda2\lib\site-packages (from httpcore==1.*->httpx<1.0,>=0.24.1->gradio) (0.14.0)

Requirement already satisfied: filelock in c:\users\suras\anaconda2\lib\site-packages (from huggingface-hub>=0.28.1->gradio) (3.13.1)

Requirement already satisfied: requests in c:\users\suras\anaconda2\lib\site-packages (from huggingface-hub>=0.28.1->gradio) (2.32.3)

Requirement already satisfied: tqdm>=4.42.1 in c:\users\suras\anaconda2\lib\site-packages (from huggingface-hub>=0.28.1->gradio) (4.66.5)

Requirement already satisfied: annotated-types>=0.6.0 in c:\users\suras\anaconda2\lib\site-packages (from pydantic<2.12,>=2.0->gradio) (0.6.0)

Requirement already satisfied: pydantic-core==2.33.2 in c:\users\suras\anaconda2\lib\site-packages (from pydantic<2.12,>=2.0->gradio) (2.33.2)

Requirement already satisfied: typing-inspection>=0.4.0 in c:\users\suras\anaconda2\lib\site-packages (from pydantic<2.12,>=2.0->gradio) (0.4.1)

Requirement already satisfied: six>=1.5 in c:\users\suras\anaconda2\lib\site-packages (from python-dateutil>=2.8.2->pandas) (1.16.0)

Requirement already satisfied: click>=8.0.0 in c:\users\suras\anaconda2\lib\site-packages (from typer<1.0,>=0.12->gradio) (8.1.7)

Requirement already satisfied: shellingham>=1.3.0 in c:\users\suras\anaconda2\lib\site-packages (from typer<1.0,>=0.12->gradio) (1.5.4)

Requirement already satisfied: rich>=10.11.0 in c:\users\suras\anaconda2\lib\site-packages (from typer<1.0,>=0.12->gradio) (13.7.1)
Requirement already satisfied: colorama in c:\users\suras\anaconda2\lib\site-packages (from click>=8.0.0->typer<1.0,>=0.12->gradio) (0.4.6)
Requirement already satisfied: markdown-it-py>=2.2.0 in c:\users\suras\anaconda2\lib\site-packages (from rich>=10.11.0->typer<1.0,>=0.12->gradio) (2.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in c:\users\suras\anaconda2\lib\site-packages (from rich>=10.11.0->typer<1.0,>=0.12->gradio) (2.15.1)
Requirement already satisfied: charset-normalizer<4,>=2 in c:\users\suras\anaconda2\lib\site-packages (from requests->huggingface-hub>=0.28.1->gradio) (3.3.2)
Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\suras\anaconda2\lib\site-packages (from requests->huggingface-hub>=0.28.1->gradio) (1.26.20)
Requirement already satisfied: mdurl~=0.1 in c:\users\suras\anaconda2\lib\site-packages (from markdown-it-py>=2.2.0->rich>=10.11.0->typer<1.0,>=0.12->gradio) (0.1.0)
Note: you may need to restart the kernel to use updated packages.

```
In [7]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Load Titanic Dataset
url = r"C:\Users\suras\Downloads\titanic_dataset_final.csv"
df = pd.read_csv(url)
df
```

Out[7]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.25
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.28
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.92
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.10
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.05
...	...	...	...	...	...	...	...	...	...	...
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.00
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.00
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.45
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.00
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7.73

891 rows × 12 columns



```
In [8]: # Display dataset info
print(df.describe())
```

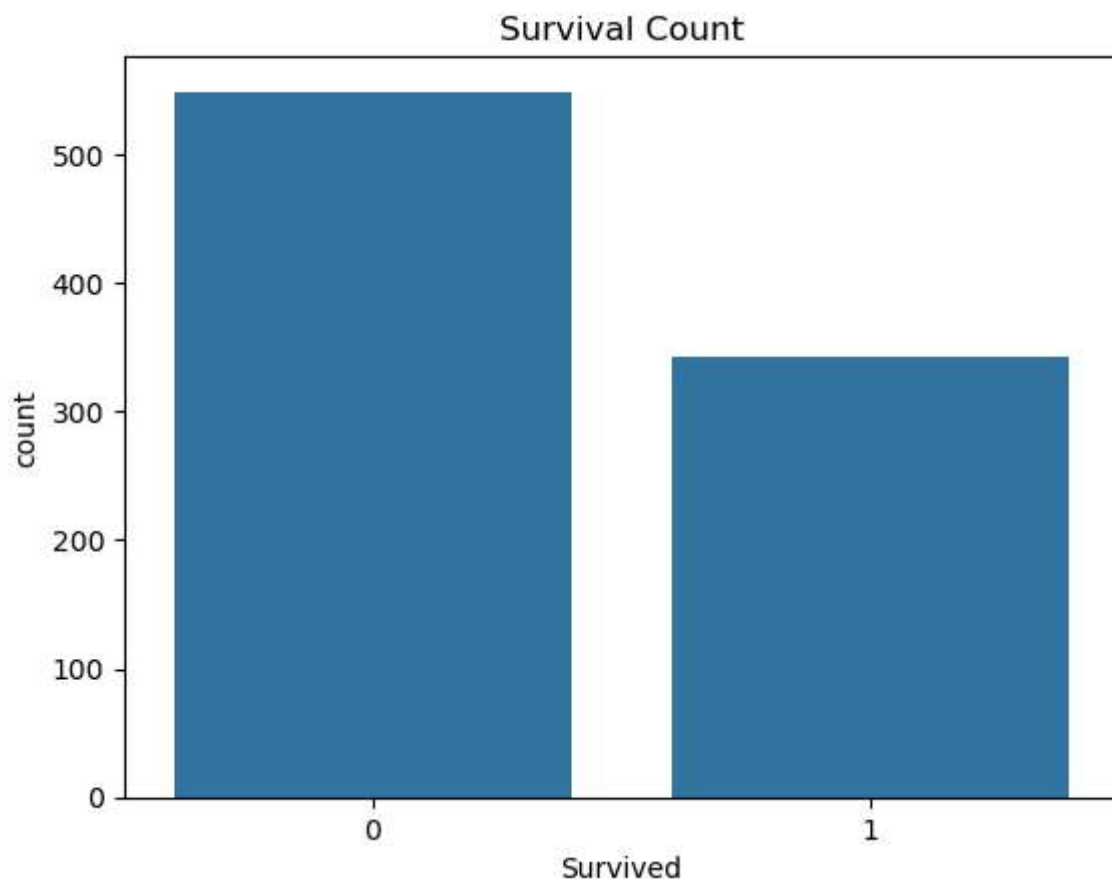
	PassengerId	Survived	Pclass	Age	SibSp \
count	891.000000	891.000000	891.000000	714.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008
std	257.353842	0.486592	0.836071	14.526497	1.102743
min	1.000000	0.000000	1.000000	0.420000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000
50%	446.000000	0.000000	3.000000	28.000000	0.000000
75%	668.500000	1.000000	3.000000	38.000000	1.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000

	Parch	Fare
count	891.000000	891.000000
mean	0.381594	32.204208
std	0.806057	49.693429
min	0.000000	0.000000
25%	0.000000	7.910400
50%	0.000000	14.454200
75%	0.000000	31.000000
max	6.000000	512.329200

```
In [9]: # Missing Values Check
print("\nMissing Values:\n", df.isnull().sum())
```

```
Missing Values:
PassengerId      0
Survived          0
Pclass           0
Name             0
Sex              0
Age             177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin           687
Embarked         2
dtype: int64
```

```
In [10]: # Survival Rate Visualization
sns.countplot(x='Survived', data=df)
plt.title("Survival Count")
plt.show()
```



```
In [14]: import ollama

def generate_insights(df_summary):
    prompt = f"Analyze the dataset summary and provide insights:\n\n{df_summary}"
    response = ollama.chat(model="mistral", messages=[{"role": "user", "content": p
    return response['message']['content']

# Generate AI Insights
summary = df.describe().to_string()
insights = generate_insights(summary)
print("\n ♦ AI-Generated Insights:\n", insights)
```

```

-----
ResponseError                                Traceback (most recent call last)
Cell In[14], line 10
      8 # Generate AI Insights
      9 summary = df.describe().to_string()
--> 10 insights = generate_insights(summary)
     11 print("\n ♦ AI-Generated Insights:\n", insights)

Cell In[14], line 5, in generate_insights(df_summary)
      3 def generate_insights(df_summary):
      4     prompt = f"Analyze the dataset summary and provide insights:\n\n{df_summary}"
--> 5     response = ollama.chat(model="mistral", messages=[{"role": "user", "content": prompt}])
      6     return response['message']['content']

File ~\anaconda2\Lib\site-packages\ollama\_client.py:342, in Client.chat(self, model, messages, tools, stream, think, format, options, keep_alive)
     297 def chat(
     298     self,
     299     model: str = '',
    (...)
     307     keep_alive: Optional[Union[float, str]] = None,
     308 ) -> Union[ChatResponse, Iterator[ChatResponse]]:
     309     """
     310     Create a chat response using the requested model.
     311     (...)
     340     Returns `ChatResponse` if `stream` is `False`, otherwise returns a `ChatResponse` generator.
     341     """
--> 342     return self._request(
     343         ChatResponse,
     344         'POST',
     345         '/api/chat',
     346         json=ChatRequest(
     347             model=model,
     348             messages=list(_copy_messages(messages)),
     349             tools=list(_copy_tools(tools)),
     350             stream=stream,
     351             think=think,
     352             format=format,
     353             options=options,
     354             keep_alive=keep_alive,
     355         ).model_dump(exclude_none=True),
     356         stream=stream,
     357     )

File ~\anaconda2\Lib\site-packages\ollama\_client.py:180, in Client._request(self, cls, stream, *args, **kwargs)
     176         yield cls(**part)
     178     return inner()
--> 180 return cls(**self._request_raw(*args, **kwargs).json())

File ~\anaconda2\Lib\site-packages\ollama\_client.py:124, in Client._request_raw(self, *args, **kwargs)

```



```
122     return r
123 except httpx.HTTPStatusError as e:
--> 124     raise ResponseError(e.response.text, e.response.status_code) from None
125 except httpx.ConnectError:
126     raise ConnectionError(CONNECTION_ERROR_MESSAGE) from None
```

ResponseError: model requires more system memory (5.7 GiB) than is available (4.2 GiB) (status code: 500)

```
In [15]: import gradio as gr

def eda_analysis(file):
    df = pd.read_csv(file.name)
    summary = df.describe().to_string()
    insights = generate_insights(summary)
    return insights

# Create Web Interface
demo = gr.Interface(fn=eda_analysis, inputs="file", outputs="text", title="AI-Power

# Launch App
demo.launch(share=True) # Use share=True for Google Colab
```

* Running on local URL: <http://127.0.0.1:7869>

* Running on public URL: <https://abd55e6acf4ee4d9b2.gradio.live>

This share link expires in 1 week. For free permanent hosting and GPU upgrades, run ``gradio deploy`` from the terminal in the working directory to deploy to Hugging Face Spaces (<https://huggingface.co/spaces>)

Out[15]:

In []: